SOURCE: MEPC.376(80).pdf
management (esca) are considered as zero for the establishment of the initial default emission factors. Similarly, this is the case also for the parameters related to carbon capture and storage (ccs), which require further development.

9.5 TtW default emission factors, including slip factors per fuel type and per converter types, are set out in appendix 2 (for those fuels and converters for which such factors are available in resolution MEPC.364(79) on the 2022 Guidelines on the method of calculation of the attained energy efficiency design index (EEDI) for new ships and the Fourth IMO GHG Study 2020). Further TtW default emission factors (with the exception of C fCO2 provided in resolution MEPC.364(79)) may be established by following the same rules as for the WtT default emission factors described in paragraph 9.3. No default emission factors are provided for the use of on board CCS (eoccs), and the amount of captured carbon per unit of fuel mass should be specifically certified. T he parameters related to emission credits from the used captured CO2 as carbon stock to produce synthetic fuels (eccu) requires further development.

---CHUNK---

SOURCE: MEPC.376(80).pdf
should be specifically certified. T he parameters related to emission credits from the used captured CO2 as carbon stock to produce synthetic fuels (eccu) requires further development.

9.6 As the definition of Cfug factors is considered a difficult parameter to be measured, Cfug factors should be established by the best existing knowledge and will be dealt with at a later stage. Until Cfug factors are defined, Cfug should be set as 0.

9.7 In case additional categories of energy converters (not listed in appendix 2) are proposed, the rules to establish TtW default emission factors as described in paragraph 9.5 above may be followed to ensure that these new converters (e.g. fuel cells) ma y also be associated with a default emission factor.

10 ACTUAL EMISSION FACTORS

10.1 The aim of actual emission factors is to allow demonstration of superior GHG performance compared to the default emission factors, subject to verification and certification by a third party.

10.2 WtT and TtW emission factors should be based on methodologies established in these guidelines. Actual values provide the WtW (WtT and TtW) GHG intensity for the specific

---CHUNK---

SOURCE: MEPC.376(80).pdf
by a third party.

10.2 WtT and TtW emission factors should be based on methodologies established in these guidelines. Actual values provide the WtW (WtT and TtW) GHG intensity for the specific fuel over the life cycle (from fuel production to its use on board).

10.3 For the pathways contained in appendix 1, the description and the calculation method for providing WtT actual emission factors should be provided. In addition, for the pathways not contained in appendix 1, a detailed description of the pathway should be provided.

10.4 The use of actual WtT emission factors is not applicable to purely fossil pathways. However, for fuels which are produced from captured carbon of fossil origin and for fossil fuels where the technology of CCS/CCUS is applied, actual values are allowed. For the fossil component of a blended fuel, fossil fuel default emission factors should be used.
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 21

I:\MEPC\80\MEPC 80-17-Add.1.docx

---CHUNK---

SOURCE: MEPC.376(80).pdf
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 21

I:\MEPC\80\MEPC 80-17-Add.1.docx
10.5 Actual TtW emission factors are allowed for all fuel pathways18 and provided in these guidelines.

PART IV: VERIFICATION AND CERTIFICATION

11 ELEMENTS SUBJECT TO VERIFICATION/CERTIFICATION

11.1 When used as evidence for performances, the FLL needs to be verified and certified by a third party, taking into account further guidance to be developed by the Organization.

11.2 The verification and certification of Part A, Part B, Part C, and Part E of the FLL may be carried out separately by different verification bodies. The verification and certification of Part D of the FLL needs to be based on the verified Part A, Part B and Part C.

11.3 For fuel types with a specific fuel pathway code and which will be consumed in a specified energy converter, the default emission factors for Part A -5, Part C-1, Part C-2 and Part D of the FLL are provided in appendix 2. As long as Part A-1 to Part A-4 and Part C-3 of

---CHUNK---

SOURCE: MEPC.376(80).pdf
specified energy converter, the default emission factors for Part A -5, Part C-1, Part C-2 and Part D of the FLL are provided in appendix 2. As long as Part A-1 to Part A-4 and Part C-3 of the FLL have been duly verified, the default emission factors contained in these guidelines can be consequently applied without further verification.

11.4 In the case where lower emission factors are claimed compared to the default emission factors for Part A-5, Part C-1, Part C-2 and/or Part D, the actual emission factors can be used only after the verification and certification by a third party, taking int o account further guidance referred to in paragraph 11.1.

12 IDENTIFICATION OF CERTIFICATION SCHEMES/STANDARDS

12.1 The verification and certification of individual parts of the FLL will use relevant

certification schemes/standards. Different parts of the FLL may be verified using different certification schemes/standards as applicable, while a specific part of the FLL may be addressed by multiple certification schemes/standards with similar scopes.

12.2 The certification schemes/standards used for the purposes specified in

---CHUNK---

SOURCE: MEPC.376(80).pdf
certification schemes/standards as applicable, while a specific part of the FLL may be addressed by multiple certification schemes/standards with similar scopes.

12.2 The certification schemes/standards used for the purposes specified in paragraph 12.1 above should be recognized by the Committee, taking into account guidance to be developed by the Organization. The list of recognized certification schemes/standards should be publicly available and kept under review.

12.3 Proposals to recognize international certification schemes/standards should be submitted to the Committee for consideration, including an assessment of a set of predetermined criteria which will be further developed for this purpose.

12.4 The framework, criteria and procedures leading to the recognition of certification schemes should be implemented uniformly to guarantee the quality, reliability and robustness of the IMO framework as a whole and to ensure a level playing field among c ertification schemes.

18 Verification and certification methodologies would need further work to be established.
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)

---CHUNK---

SOURCE: MEPC.376(80).pdf
schemes.

18 Verification and certification methodologies would need further work to be established.
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 22

I:\MEPC\80\MEPC 80-17-Add.1.docx
PART V: REVIEW

13 CONTINUOUS REVIEW PROCESS

13.1 To ensure that new technological advances and scientific knowledge are taken into account, these guidelines should be kept under continuous technical review taking into account emerging and evolving technologies.

13.2 In particular, the following elements should be kept under review:

.1 WtT, TtW and WtW default emission factors as specified in appendix 2; and

.2 new proposed fuel pathways and the corresponding default emission factors
in addition to those specified in appendix 1.

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 23

I:\MEPC\80\MEPC 80-17-Add.1.docx
APPENDIX 1

FUEL LIST WITH FUEL PATHWAY CODES

Order Group Fuel type

---CHUNK---

SOURCE: MEPC.376(80).pdf
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 23